

Thermal Physics

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1 Introduction

These are my notes from **Thermal Physics by Katherine Blundell and Stephen Blundell**

Example 1.1: 1 kg of N_2 gas $\rightsquigarrow 2 \times 10^{25}$ N_2 molecules.

- 1 year = 3.2×10^7 seconds.
- A 3 GHz personal computer can count 10^{17} per year.
- It would take roughly 0.2 billion years just to count the molecules in 1 kg of gas.

Since we cannot track every molecule, we make **approximations** and deal with **statistical properties** (averages).

1.1 What is a Mole?

A mole is the quantity of matter that contains as many objects as the number of atoms in exactly 12 g of ^{12}C .

$$N_A = 6.022 \times 10^{23}$$

Example 1.2:

- 1 mole of C = 6.022×10^{23} atoms of C.
- 1 mole of benzene = 6.022×10^{23} molecules of benzene.
- 1 mole of NaCl = 6.022×10^{23} NaCl formula units.

**Note: A mole of eggs would make an omelette with about half the mass of the Moon!*

1.2 Molar Mass

Molar mass (mass/mole) for carbon is 12 g, and for water is 18 g (which means it is $\frac{18}{12}$ times the mass of a carbon atom).

$$m = \frac{\text{molar mass}}{N_A} \Rightarrow \text{molar mass} = mN_A$$

2 Thermodynamic Limit

Imagine the rooftop of a small hut in rainfall. Rain hitting the roof can be graphed as an impulse graph at each point. As the roof gets bigger, this impulse gets flatter (less fluctuation). Fluctuations grow slower than the force of impact as area $\rightarrow \infty$.

Pressure is defined as:

$$P = \frac{\text{Force}}{\text{Area}}$$

This limit is called the **thermodynamic limit**: a container with an infinite number of molecules exerting uniform pressure.

2.1 Variables

- **Extensive Variables:** Change with size (e.g., Kinetic Energy U , Volume V).
- **Intensive Variables:** Do not change with size (e.g., Temperature T , Pressure P).

3 The Ideal Gas

- Const. $T \implies P \propto \frac{1}{V}$ (Boyle's Law)
- Const. $P \implies V \propto T$ (Charles's Law)
- Const. $V \implies P \propto T$ (Gay-Lussac's Law)

For N molecules in a gas:

$$PV = Nk_B T \quad (\text{Ideal Gas equation*})$$

Where k_B (Boltzmann constant) $\approx 1.3807 \times 10^{-23} \text{ J K}^{-1}$.

4 Combinatorial Problems

Example 1.3: Arranging r -quanta among n -atoms.

$$\Omega_{\text{guess}} = n \times (n-1) \times (n-2) \times \cdots \times (n-r+1) = \frac{n!}{(n-r)!}$$

Since quanta are indistinguishable, we divide by $r!$ for unique arrangements:

$$\Omega = \frac{n!}{(n-r)!r!} \equiv \binom{n}{r} \quad (\text{Combination})$$

4.1 Stirling's Formula

Logarithms make larger numbers manageable:

$$\ln \Omega = \ln(n!) - \ln((n-r)!) - \ln(r!)$$

Approximation for large n :

$$\ln(n!) \approx n \ln(n) - n$$

Derivation of Stirling's Formula

We start with the integral representation of the Gamma function for $n!$:

$$n! = \int_0^\infty x^n e^{-x} dx$$

Let the integrand be represented as $e^{f(x)}$:

$$e^{f(x)} = x^n e^{-x} \implies f(x) = n \ln x - x$$

To find the maximum of $f(x)$, we take the first derivative:

$$\frac{df(x)}{dx} = \frac{n}{x} - 1 = 0 \implies x = n$$

The second derivative at the maximum $x = n$ is:

$$\left. \frac{d^2 f(x)}{dx^2} \right|_{x=n} = -\frac{n}{x^2} \Big|_{x=n} = -\frac{1}{n}$$

Expanding $f(x)$ as a Taylor series around $x = n$:

$$\begin{aligned} f(x) &\approx f(n) + f'(n)(x-n) + \frac{1}{2!} f''(n)(x-n)^2 + \dots \\ &\approx (n \ln n - n) + 0 + \frac{1}{2} \left(-\frac{1}{n} \right) (x-n)^2 \\ &= n \ln n - n - \frac{(x-n)^2}{2n} \end{aligned}$$

Substituting this back into the integral for $n!$:

$$n! \approx \int_0^\infty e^{n \ln n - n} \cdot e^{-\frac{(x-n)^2}{2n}} dx$$

Since $e^{n \ln n - n}$ is a constant relative to x , we pull it out. As $n \rightarrow \infty$, the lower limit 0 can be extended to $-\infty$ because the Gaussian decays rapidly:

$$n! \approx e^{n \ln n - n} \int_{-\infty}^\infty e^{-\frac{(x-n)^2}{2n}} dx$$

Using the standard Gaussian integral $\int_{-\infty}^\infty e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$ with $a = \frac{1}{2n}$:

$$n! \approx e^{n \ln n - n} \sqrt{2\pi n}$$

Taking the natural logarithm of both sides:

$$\ln(n!) \approx n \ln n - n + \frac{1}{2} \ln(2\pi n)$$

For very large n , the first terms dominate, yielding the simple form:

$$\ln(n!) \approx n \ln n - n$$